SE444 Mock Final Exam – Answer Key & Grading Guide (Fall 2025)

Part I: Multiple Choice (6 pts)

Option	Q1 (1pt)	Q2 (1pt)	Q3 (1pt)	Q4 (1pt)	Q5 (1pt)	Q6 (1pt)
A	X		X	X	X	X
B		X	X		X	
C		X	X	X		X
D	X	X	X	X	X	
E				X	X	X

Answers: Q1: A,D — Q2: B,C,D — Q3: A,B,C,D — Q4: A,C,D,E — Q5: A,B,D,E — Q6: A,C,E
Grading: 1pt if all correct; 0.5pt if 1 error; 0pt if 2+ errors.

Part II: Probability & BN (11 pts)

Q II.1: BN + d-Separation (4 pts)

(a) BN Structure [2 pts]:
- $I \to N$, $S \to N$ (both cause traffic)
- $I \to F$, $N \to F$ (firewall depends on both)
- $I \to L$, $S \to L$ (log depends on both)
- Colliders: N and L

(b) $I \perp S \mid F$**?** **[1 pt]:** **NO** – Observing F activates path through collider N (since $N \to F$).

(c) Explaining Away [1 pt]: Collider = N. If high traffic observed, learning S=high "explains away" the traffic, reducing belief in I.

Q II.2: Bayesian Inference (4 pts)

(a) $P(H = T|L = T)$ **[2 pts]:**

$$P(L = T|H = T) = 0.9 \times 0.95 + 0.1 \times 0.05 = 0.86$$
$$P(L = T|H = F) = 0.1 \times 0.95 + 0.9 \times 0.05 = 0.14$$
$$P(L = T) = 0.86(0.6) + 0.14(0.4) = 0.572$$
$$P(H = T|L = T) = \frac{0.516}{0.572} = \mathbf{0.902}$$

(b) Variable Elimination [2 pts]: Same result via factor multiplication + marginalization over M.

Q II.3: MEU Decision (3 pts)

(a) Expected Utility [2 pts]:
- EU(Fly) $= 0.6(100) + 0.3(60) + 0.1(-50) =$ **73**
- EU(Detour) $= 0.6(70) + 0.3(70) + 0.1(30) =$ **66**
- EU(Wait) $= 0.6(50) + 0.3(50) + 0.1(80) =$ **53**

(b) MEU Action [1 pt]: **Fly Direct** (EU=73)

Part III: MDPs (5 pts)

Q III.1: MDP + Value Iteration (5 pts)

(a) MDP Components [1 pt]:
- $S = \{S1, S2, S3, S4, S5, G\}$, $A = \{U, D, L, R\}$
- P: 80% intended, 10% left, 10% right
- $R = -1$ step, $R(G) = +10$, $\gamma = 0.9$

(b) Value Iteration [2 pts]:
Iter 1: $V_1(S3) = V_1(S5) = -1 + 0.9(0.8 \times 10) =$ **6.2**
Iter 2: $V_2(S3) \approx$ **6.6**, $V_2(S5) \approx$ **7.0**

(c) Optimal Policy [2 pts]:
- $\pi^*(S3) =$ **Down** (toward G)
- $\pi^*(S5) =$ **Right** (toward G)

Part IV: Planning & Game Theory (6 pts)

Q IV.1: STRIPS Planning (3 pts)

(a) Load(pkg,truck,dock) [1 pt]:
- Pre: At(pkg,dock), TruckAt(truck,dock), Sorted(pkg)
- Add: In(pkg,truck)
- Del: At(pkg,dock)

(b) Plan [2 pts]: Sort(P1) → Move(P1,DockA) → Load(P1) → Sort(P2) → Move(P2,DockA) → Load(P2)

Q IV.2: Nash Equilibrium (3 pts)

(a) Dominated [1 pt]: Premium dominated by Budget for both players.

(b) Nash Equilibria [2 pts]:
- **(Budget, Budget) = (5,5)**
- **(Freemium, Freemium) = (7,7)**

Part V: Project Code (5 pts)

Q V.1: A* Search (3 pts)

(a) g = cost from start, h = heuristic to goal, $f = g + h$.
(b) Explored check prevents re-expanding nodes (efficiency).

(c) Yes – Manhattan is admissible (never overestimates).

Q V.2: Bayesian Updates (2 pts)

(a) Posterior [1 pt]: $P(\text{obs}|\text{det}) = \frac{0.9 \times 0.3}{0.9(0.3)+0.1(0.7)} = \frac{0.27}{0.34} =$ **0.794**

(b) Accuracy=0.5 [1 pt]: Posterior = Prior (0.3). Uninformative sensor provides no update.